SHRIKANT SHARMA

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Data Scientist | Pharma & Financial Services | Python, ML, NLP, GenAI, Agentic AI

SKILLS

Machine Learning & AI: Classification, Regression, Statistical Modeling, Causal Inference, Survival Analysis, Experiment Design, NLP, Generative AI, LLMs, Agentic AI, RAG, Deep Learning, Feature Engineering, Anomaly Detection

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Sentence Transformers, FAISS, SHAP, RAGAS, LangGraph, DoWhy, lifelines, PySpark, Matplotlib), SQL, R

Cloud & Data Platforms: AWS (S3, EC2, Lambda, ECR), Databricks, Snowflake

Tools & Platforms: Tableau, Power BI, Apache Spark, Git, Docker, FastAPI, Streamlit, MLflow

EXPERIENCE

Amgen | **Data Scientist** — **Sept 2022 – May 2026**

- Architected the Python ETL pipeline behind Amgen's CMS Open Payments submission, processing ~475K records (~$241M) on AWS and Databricks with automated SQL validation that held the failure rate under 0.5%. Designed year-over-year Tableau dashboards and led process improvements that cut processing time from 2 weeks to 4 days over 3 annual submission cycles.
- Built a multivariate risk scoring system in Python (Pandas, NumPy, Scikit-learn) that ranked 65+ countries across 20+ compliance indicators using z-score normalization. Rebalanced feature weights from 50/50 to 75/25 in favor of quantitative data and expanded the output from 3 risk tiers to 5, which surfaced 3 high-risk markets that had gone undetected. Presented findings to Directors and Executive Directors, informing resource allocation across 3 global regions.
- Developed a statistical flagging system in Python for compliance transactional monitoring across 5 payment processes (Speaker Programs, Advisory Boards, Donations, Fee-for-Service, Sponsorships), applying threshold-based detection on 50K+ records across the US, ELMAC, and JAPAC. Reduced manual review volume by 35%.
- Led an initiative to apply NLP to compliance document processing. Delivered a working text classification prototype using TF-IDF and Random Forest (Python, Scikit-learn) that cut manual categorization work by 25%. Also contributed to Amgen's enterprise AI direction through 2 internal GenAI hackathons.
- Delivered 3 concurrent projects on schedule while simultaneously owning end-to-end knowledge transfer of analytics workflows to incoming team members during organizational restructuring; mentored 3 junior data scientists on Python automation and Tableau development.

Nugget.ai | **Data Scientist Intern** — **Jan 2022 – May 2022**

- Trained classification models in Python (Scikit-learn) to predict candidate-role fit from behavioral assessments. Tested Logistic Regression, Random Forest, and Gradient Boosting with cross-validation and improved matching accuracy by 15% over the platform's existing baseline.
- Created 20+ features from raw assessment data using Pandas and NumPy, including response structure, vocabulary patterns, and completion signals, which improved model performance and informed the product team's roadmap.

Atos Syntel | **Data Scientist** | *Pune, India* | *Client: American Express* — **Feb 2018 – Dec 2019**

- Built risk scoring pipelines in Python and SQL, processing 10M+ transactions, engineering features around spending velocity, geographic patterns, and behavioral segmentation that fed downstream fraud and credit risk team reviews.
- Designed Tableau dashboards consolidating 5+ source systems into a single executive view used in the weekly credit risk review, supporting mitigation decisions totaling over $2M and cutting reporting time by 30%.
- Implemented the team's A/B testing framework in Python, SQL, and Git, handling experiment design, sample sizing, and significance testing across 12+ cycles per year. Also owned the data validation pipeline that maintained 99%+ accuracy for regulatory reporting.

Atos Syntel | **Data Analyst** | *Chennai, India* | *Client: American Express* — **Mar 2016 – Jan 2018**

- Established the SQL workflows that processed 5M+ financial transaction records every month, with validation logic and quality checks that held accuracy at 99.5% for regulatory reporting.
- Automated 15+ recurring reports using SQL and Python, freeing up 30+ hours of manual work each month so the team could focus on analysis instead of data prep.

PROJECTS

Clinical Trial Intelligence System | *Python, LangGraph, Sentence Transformers, PubMedBERT, Cross-Encoder, Groq Llama 3.3 70B, FAISS, RAGAS, Streamlit, Git* **May 2026**

github.com/Shrikant-Sharma/clinical-trial-rag | shrikant-clinical-rag.streamlit.app

- Built a clinical-trial search assistant over 484 ClinicalTrials.gov oncology protocols (3,264 chunks), so a research analyst could surface relevant studies through natural-language queries instead of manual abstract scanning.
- Compared 3 chunking strategies (500/1000/1500 chars) and evaluated retrieval with precision@k and MRR and generation with RAGAS faithfulness scores across a 25-query stress test.
- Architected agentic Corrective RAG (CRAG) upgrade with LangGraph adding LLM-as-judge document grading, query rewriting, and bounded retries. Three independent refusal gates correctly refused adversarial queries scoring 0.91 cosine similarity that the baseline threshold missed.
- Added a cross-encoder reranker (MS-MARCO MiniLM-L-6-v2) for two-stage retrieval before deduplication; halved in-corpus refusal rate (67%→33%) on a 6-query LLM-judged eval by promoting previously-missed trastuzumab-deruxtecan (T-DXd) chunks above generic trastuzumab studies on HER2-positive antibody-drug conjugate queries. Deployed live on Streamlit Community Cloud with graceful rate-limit handling.

Patient Risk Stratification Pipeline | *Python (Scikit-learn, XGBoost, PyTorch, SHAP, DoWhy, lifelines, scikit-survival), MLflow, FastAPI, Docker, AWS (Lambda, ECR, Mangum), Git* **May 2026**

github.com/Shrikant-Sharma/patient-risk-stratification | Live API on AWS Lambda

- Built an end-to-end ML pipeline for clinical heart disease risk stratification on UCI Heart Disease (303 patients, 13 clinical features); selected XGBoost from 5 model families via MLflow experiment tracking and 5-fold cross-validation, reaching 0.95 AUC-ROC and 0.93 recall (precision 0.81) at the clinical operating threshold, identifying 93 of 100 high-risk patients for early review.
- Added SHAP TreeExplainer for global and per-patient feature attributions critical for clinical decision support, validated probability calibration at Brier 0.092 with reliability curves, and tuned the decision threshold to prioritize recall while maintaining precision ≥ 0.80.
- Extended the pipeline to causal inference and survival analysis on the NHEFS cohort (1,629 subjects); estimated treatment effects via propensity score matching and G-computation with DoWhy (three-way convergence within 0.2 kg of the IPW reference, 3 sensitivity tests passing) and modeled 10-year mortality with Cox PH and Random Survival Forest at 0.80 concordance, revealing that unadjusted log-rank significance (p=0.005) was pure age confounding.
- Deployed live as a public REST API on AWS Lambda via Amazon ECR and Mangum (ASGI-to-Lambda adapter); multi-stage Docker build, EventBridge cold-start warmer holding warm latency at ~275ms, and FastAPI Swagger UI for interactive testing.

Pharma Compliance Spend Analytics | *Python (Pandas, Scikit-learn, PySpark), Snowflake, Tableau, Git* **May 2026**

github.com/Shrikant-Sharma/pharma-compliance-spend-analytics | public.tableau.com/app/profile/shrikant.sharma

- Built a compliance spend analytics system on real CMS Open Payments data (16M+ records, $13B) for federal payment-transparency review, engineering 5 HCP-level features across 989K sampled transactions and 288K providers.
- Applied within-specialty z-scores plus global IQR with a $500 monetary floor and concentration logic; flagged the top 1.67% of providers as a HIGH-tier compliance triage queue.
- Layered Isolation Forest ML anomaly detection on top; reconciliation against rule-based flags surfaced three compliance archetypes: captured specialists (BOTH), captured generalists (rules-only), and industry consultants (ML-only), revealing patterns no single method catches alone.
- Implemented the same feature pipeline across three execution backends (Pandas, Snowflake, PySpark) with row-level equivalence verified; Snowflake SQL quantified that the top 4 device manufacturers (Arthrex, Stryker, Zimmer Biomet, Smith+Nephew) capture 59.7% of orthopedic surgery payments, visualized in two interactive Tableau Public dashboards.

EDUCATION

Clark University, Worcester, MA | **Master of Science in Data Analytics**, GPA: 3.6/4.0 **May 2022**

Sant Gadge Baba Amravati University, India | **B.E. in Electronics & Telecommunication**, CGPA: 7.4/10.0

ACHIEVEMENTS

Acclaim Award (3-Star) | Amgen | For compliance analytics work and cross-functional collaboration **Feb 2025**

INTERESTS

Hiking state parks with my dog, skiing, trance music (Armin van Buuren), and strength training.